

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460967.777 +/- 0.0066 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.208 +/- 0.043
Transit depth (Rp/Rs)^2	4.31 +/- 1.8 [%]
Orbital Inclination (inc)	84.26 +/- 2.17
Airmass coefficient 1 (a1)	1.038 +/- 0.011
Airmass coefficient 2 (a2)	-0.03 +/- 0.14
Scatter in the residuals of the lightcurve fit is	1.09 %
Best Comparison Star	#3 - [283, 399]
Optimal Aperture	4.45
Optimal Annulus	5.93
Transit Duration (day)	0.041 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.208 ± 0.043 vs 0.1646 (deviation 0.96σ)
Duration fit vs published	0.041 d vs 0.0758 d (deviation 2.37σ)
Mid-time O–C	-10.6 min
Depth SNR	4.6
Residual scatter	1.09 %

Light curve

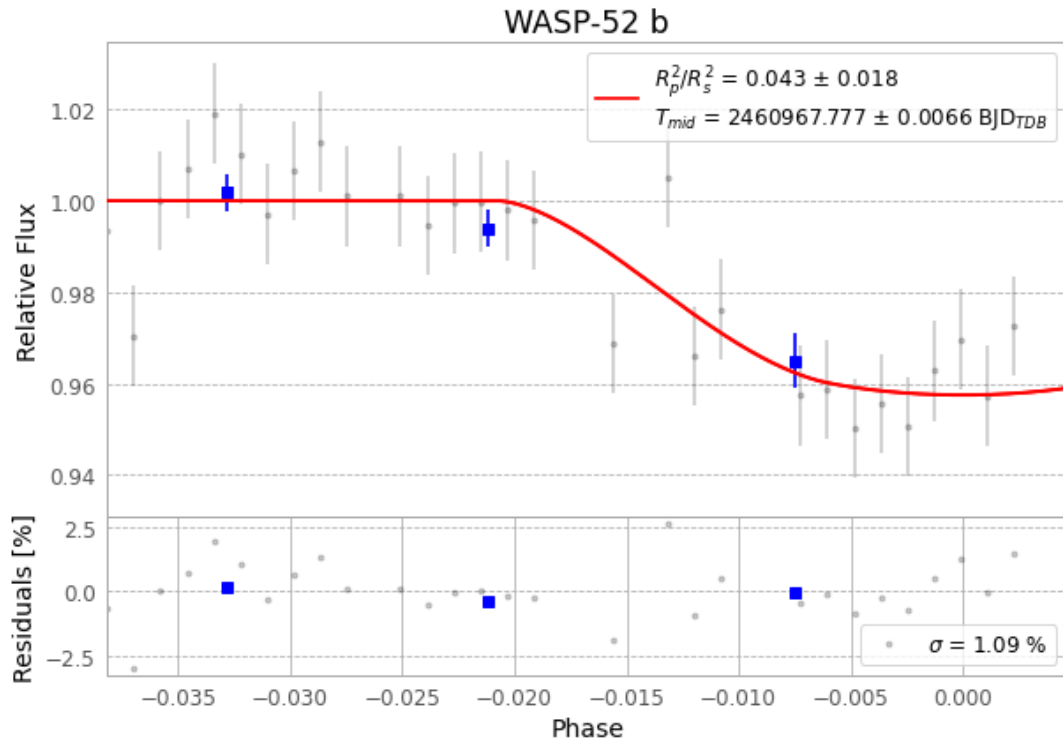


Figure 1 — FinalLightCurve_WASP-52 b_2025-10-18.png (SHA-256 fd6965cfca6e95b1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [307, 329], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 138295e3b75308d3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit faa81f0

Data provenance

32 FITS frames (21 MB), WASP-52251019045415.FITS → WASP-52251019064214.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-251019.log (1d6bd0a8b4d1...), FinalLightCurve_WASP-52 b_2025-10-18.png (fd6965cfca6e...), FinalLightCurve_WASP-52 b_2025-10-18.csv (cfae898a8163...)

Compute resources

Wall time 130.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 16:00Z.